Advanced Programming Practice · 21CSC203P

Unit 1 · Session 12 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 12 : SLO – 1 | Dynamic Dispatch & Overhead

Activity (Solutions)

• 1) class Base { void show(){ System.out.println("Base"); } } class Derived extends Base { @Override void show(){ System.out.println("Derived"); } } Base b = new Derived(); b.show(); → prints Derived (dynamic dispatch via base reference).

• 2) Virtual method table (vtable): each class with overridable methods has a table mapping method slots to implementations; an object stores a pointer to its class's vtable. For Base b = new Derived(); b.show(); the JVM reads Derived's vtable entry for show() and jumps to Derived.show — this is how overriding is resolved at run time.

• 3) Direct vs virtual call: a direct (static/private/final) call binds at compile time — the address is fixed, so it is faster and inlinable. A virtual call must load the vtable pointer and index the table at run time — a small overhead (extra indirection, harder to inline), in exchange for polymorphic flexibility.